

Mutual Fund Analytics & Investor Insights

Bluestock Fintech Capstone Project

Submitted By:

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Technologies Used:

Python • SQLite • Power BI • GitHub • VS Code

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Problem Statement & Objectives

Problem Statement

- Mutual fund data is scattered across multiple sources.
- Investors struggle to compare fund performance and risk.
- Manual analysis is time-consuming.

Objectives

- Build an ETL pipeline
- Perform Exploratory Data Analysis
- Calculate financial performance metrics
- Build Power BI Dashboard
- Develop Mutual Fund Recommendation System

Data Sources

Datasets Used

- Fund Master
- NAV History
- Scheme Performance
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices
- Industry Statistics

Technologies

- Python
- Pandas
- SQLite
- Power BI
- Git & GitHub

Project Architecture (ETL)

Insert this diagram using **SmartArt → Process**

Raw CSV Files



Data Cleaning



Data Transformation



SQLite Database



Power BI Dashboard



Reports & Insights

Bottom of slide

Tools

Python → Pandas → SQLite → Power BI

EDA Highlights (Part 1)

Insert 4 charts:

- AUM Growth
- Folio Growth
- Gender Distribution
- Age Group Distribution
- Bottom:

Key Insights

Industry AUM increased steadily.

Folio count increased every year.

Most investors belong to working-age groups.

Investor participation continues to grow.

EDA Highlights (Part 2)

Insert:

- State-wise Investment
- Sector Allocation
- Category Heatmap
- T30 vs B30

Bottom:

Insights

- Maharashtra and Karnataka contributed high investments.
- Equity sector dominated portfolios.
- Smaller cities showed increasing participation.
- Investment categories showed clear diversification.

Performance Analytics

Financial Metrics Calculated

- Daily Returns
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Key Result

Risk-adjusted metrics helped identify the best-performing mutual funds.
(Optional: Insert Rolling Sharpe chart.)

Advanced Analytics

Advanced Features

- Historical VaR (95%)
- CVaR
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- HHI Concentration Analysis
- Mutual Fund Recommendation System

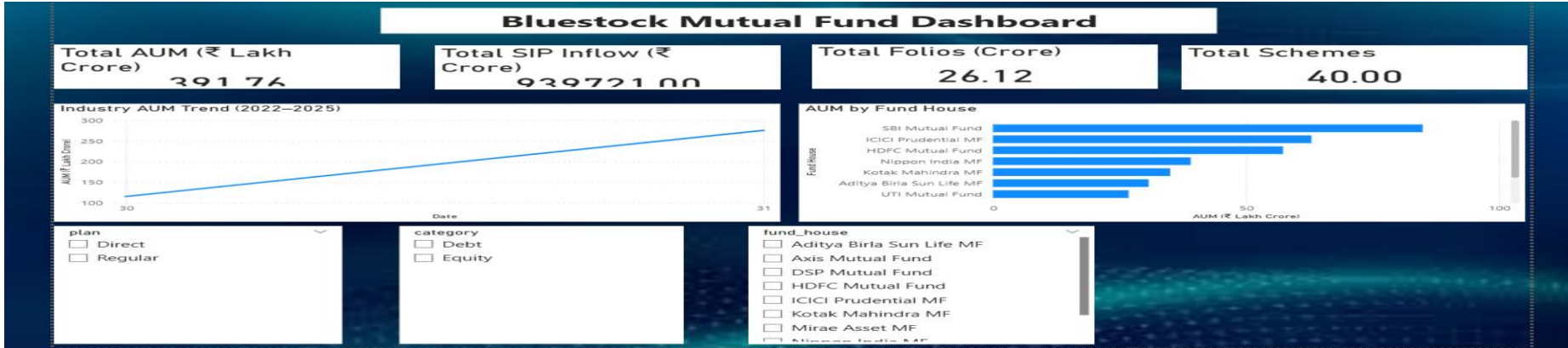
Outcome

- Investors receive risk-based fund recommendations.

Power BI Dashboard (Pages 1 & 2)

Insert screenshots of:

1. Industry Overview



2. Fund Performance



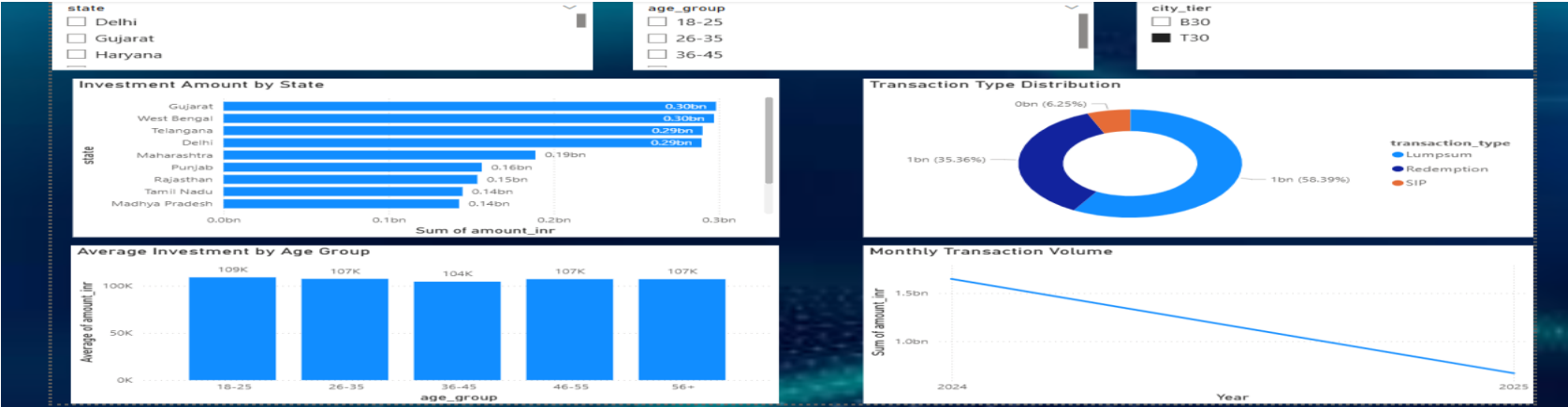
Dashboard Features

- KPI Cards
- AUM Trend
- Risk vs Return Scatter
- Fund Scorecard
- Interactive Filters

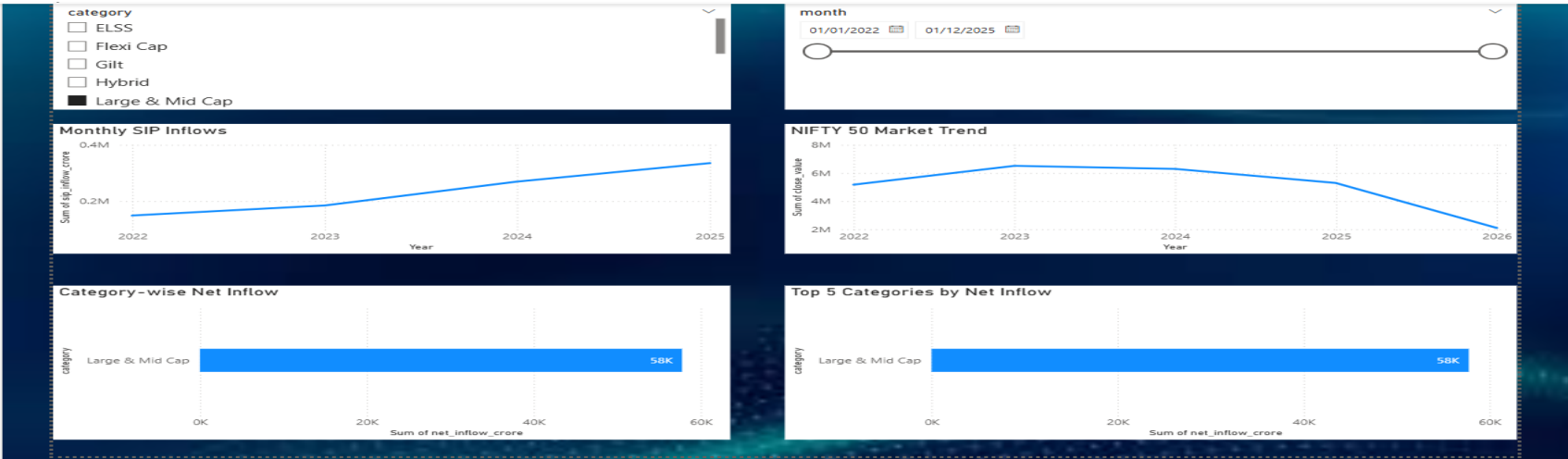
Power BI Dashboard (Pages 3 & 4)

Insert screenshots of:

3. Investor Analytics



4. SIP & Market Trends



Dashboard Features

- State Analysis
- SIP Distribution
- Monthly Transactions
- Market Trend Analysis
- Drill-through
- Tooltips

Key Findings

Major Findings

- Mutual fund industry experienced consistent growth.
- SIP inflows increased steadily.
- Risk-adjusted metrics improved fund comparison.
- Investor cohorts showed changing investment behavior.
- Power BI dashboard enabled interactive analysis.
- Recommendation system suggested top-performing funds based on risk.

Thank You

Questions?

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Thank You!